MIL-M-28787/29B
19 July 1982
SUPERSEDING
MIL-M-28787/29A(EC)
21 September 1979

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

INVERTER, DIGITAL, KEY CODE FDH

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic inverter module.

1.2 Classification. This specification covers one class of modules.

Class 1 as specified in MIL-STD-1389 except as modified in 3.4.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5963

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 2.05 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389 except the operating temperature range shall be 0° to 80°C.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be FDH.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A 40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/29-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table 1.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table 1 (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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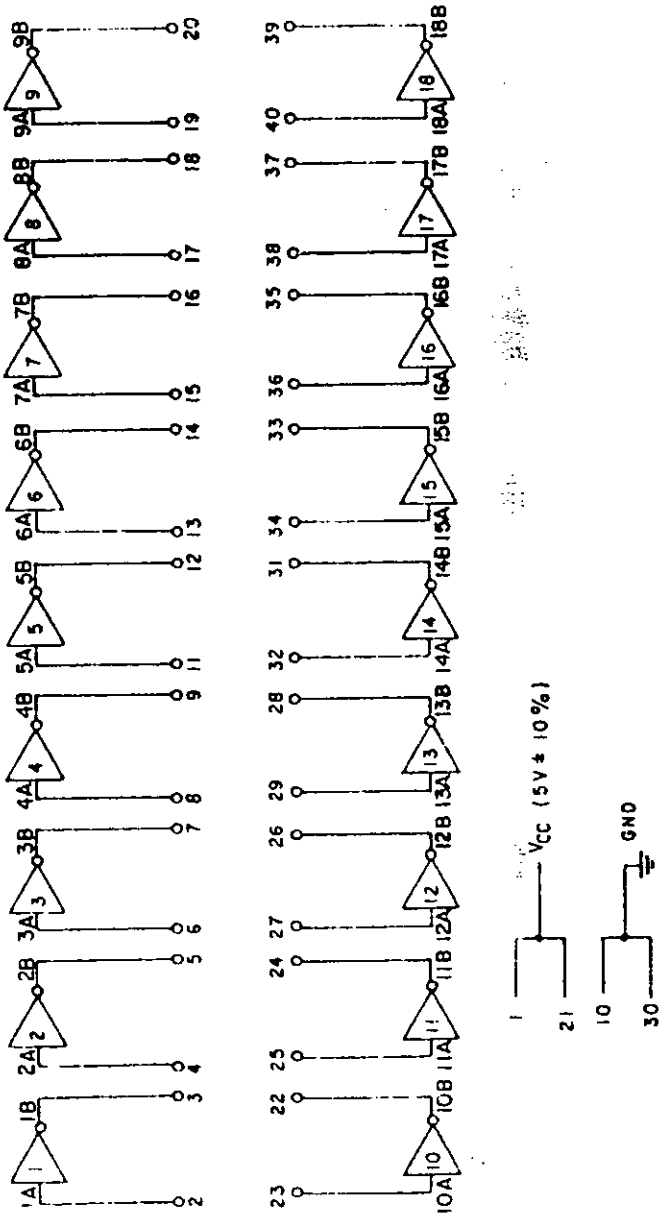


FIGURE 1. FDH module function diagram, inverter.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 80°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A18	V_{OUT} (low loaded)	III	S	0.1	0.8	0.1	0.8	0.1	0.8	0.1	0.8	V
B1-B18	Static input current	IV	S	0	500	0	500	0	500	0	500	µA
C1-C18	Input clamp diode voltage	V	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D1-D18	Output voltage, low level	VI	X	0	450	0	450	0	450	0	450	mV
E1-E18	Output voltage, high level	VI	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-F18	Input current, low level	VII	X	1.0	2.2	1.0	2.2	1.0	2.2	1.0	2.2	mA
G1-G18	Input current, high level	VII	X	0	60	0	60	0	60	0	60	µA
H1-H18	Output short circuit current	VII	X	40	100	35	100	35	100	25	100	mA
J1	Power supply current, low level	VIII	S	25	180	25	180	25	180	25	180	mA
J2	Power supply current, high level	VIII	S	10.0	75.6	10.0	75.6	10.0	75.6	10.0	75.6	mA
K1-K18	Transition time (t_{THL})	IX	S	3	20	3	20	3	25	3	25	ns
L1-L18	Transition time (t_{TLH})	IX	S	3	20	3	20	3	25	3	25	ns
M1-M18	Propagation delay time (t_{PLH})	X	S	3	20	3	20	3	25	3	25	ns
N1-N18	Propagation delay time (t_{PHL})	X	S	3	20	3	20	3	25	3	25	ns
P	Power supply transient amplidude	XI	Q	0	400	0	400	0	400	0	400	mV
Q	Isolation	XII	S	10	∞	10	∞	10	∞	10	∞	MΩ
R1-R18	Output crosstalk, low level	XIII	Q	0	400	0	400	0	400	0	400	mV
S1-S18	Output crosstalk, high level	XIII	Q	0	400	0	400	0	400	0	400	mV
T1-T18	Input capacitance	XIV	Q	2	12	2	12	2	12	2	12	pF
U	Power supply transient rise time	XV	S	2.5	7.5	2.5	7.5	2.5	7.5	2.5	7.5	µs
V1,V2	Continuity test	XVI	X	0	1	0	1	0	1	0	1	Ω

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.6 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through S, U, and V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through S, U, and V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through V shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military Specification MIL-M-28787/29 supersedes Department of the Navy, Naval Ordnance System Command Drawing No. 10001-2658751. Part number M28787/29-1 supersedes part numbers 2658751 and M28787/29.

6.4 Module description. The following electrical parameters reflect the 0° to 80°C end-of-life limits from table I.

Supply voltage	5 V dc ±10%
Maximum power dissipation	
Low level	990 mW (outputs not loaded)
High level	415.8 mW (outputs not loaded)

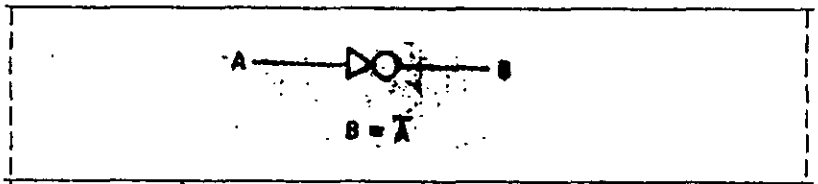
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Input voltage	
Low level	0.8 V maximum
High level	2.0 V minimum
Input current	
Low level (V_{IL} = 0 V)	2.2 mA maximum
High (V_{IH}) = 2.4 V)	60 µA maximum
Output voltage	
Low level	0.45 V maximum
High level	2.4 V minimum, V_{CC} maximum
Output drive current	
Low level (V_{OL} = 0 V)	22 mA maximum
High level (V_{OH} = 2.4 V)	0.6 mA maximum

6.5 Module function. This module contains 18 inverter gates with independent inputs and outputs.

6.5.1 Input to output relationship. Each inverter gate has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.



Input	Output
Low level	High level
High level	Low level

6.5.2 Parallel outputs. Parralleling of outputs is not permissible.

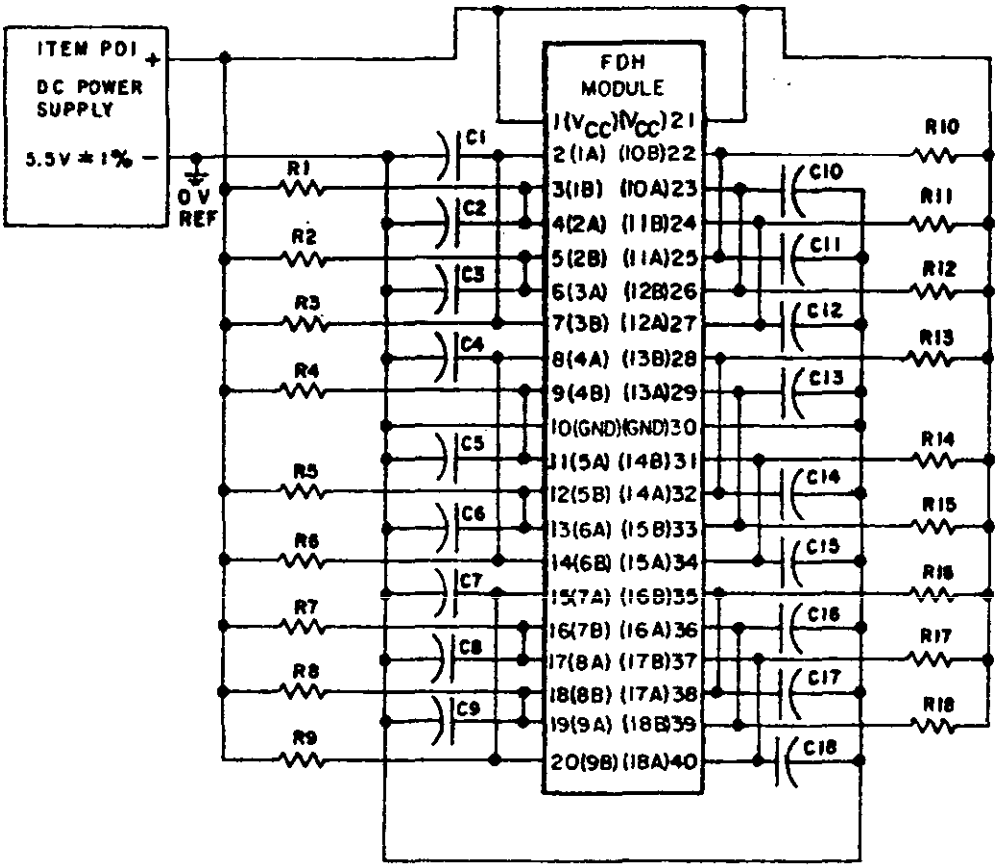
6.6 Changes from previous issue. Asterisks are not used in the revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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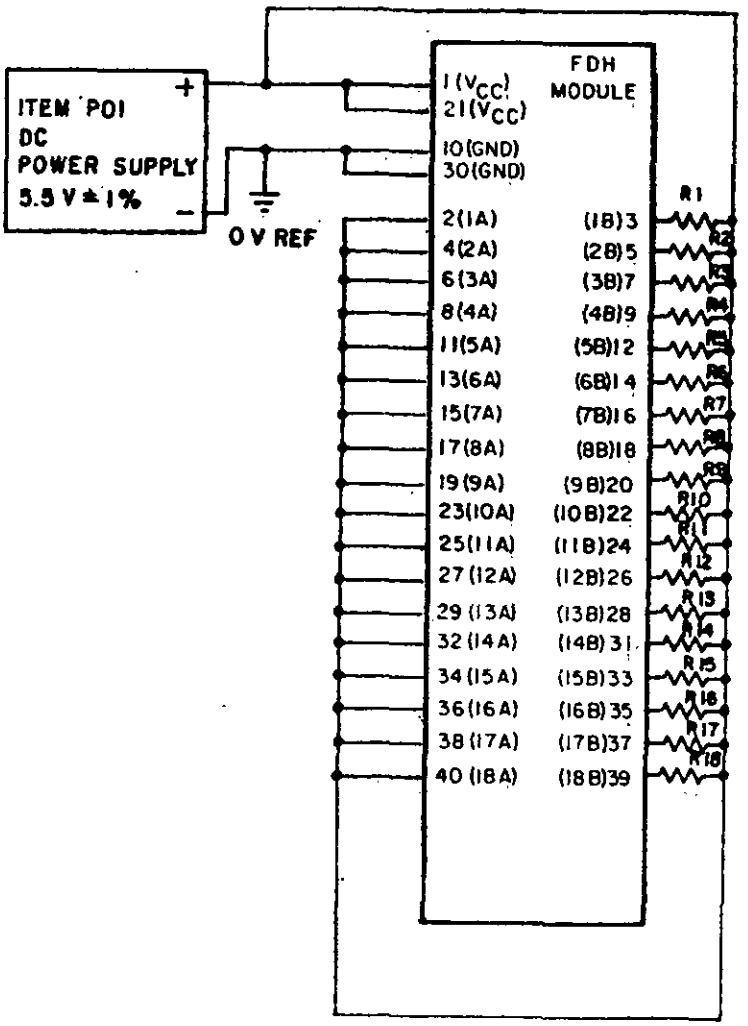
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NOTES:
1. R1-R18 = 243Ω ±1%, 1/4 W.
2. C1-C18 = 130 pF ±20%.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R18 = 243 Ω, 1/8 W.

FIGURE 3. Component [illegible] - Sheet 3.

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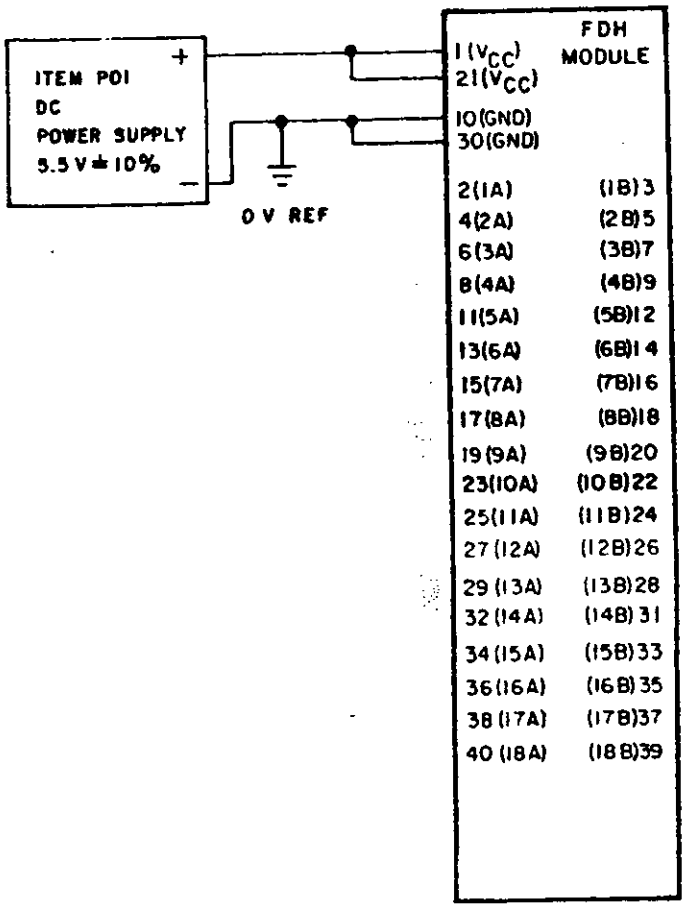


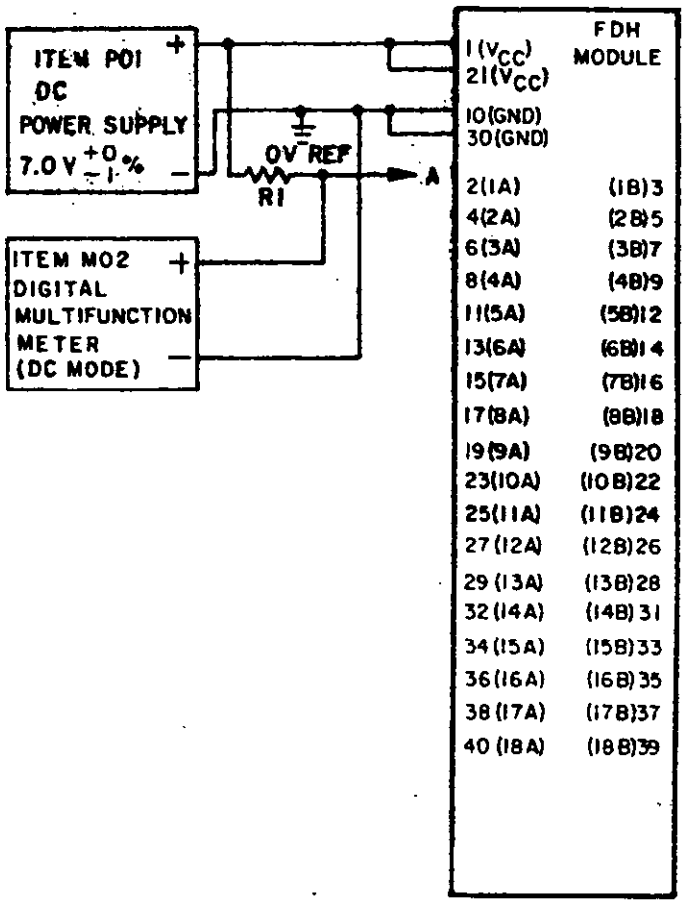
FIGURE 4. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
A1	V_{OUT} (low loaded)	3	0.1	0.8	V	5
A2		5				
A3		7				
A4		9				
A5		12				
A6		14				
A7		16				
A8		18				
A9		20				
A10		22				
A11		24				
A12		26				
A13		28				
A14		31				
A15		33				
A16		35				
A17		37				
A18		39				

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NOTES:
1. R1 = 243Ω ±1%, 1/4 W.
2. After point A has been connected to the appropriate module pin, the output voltage shall be monitored for a minimum of five seconds before the measurement is recorded.

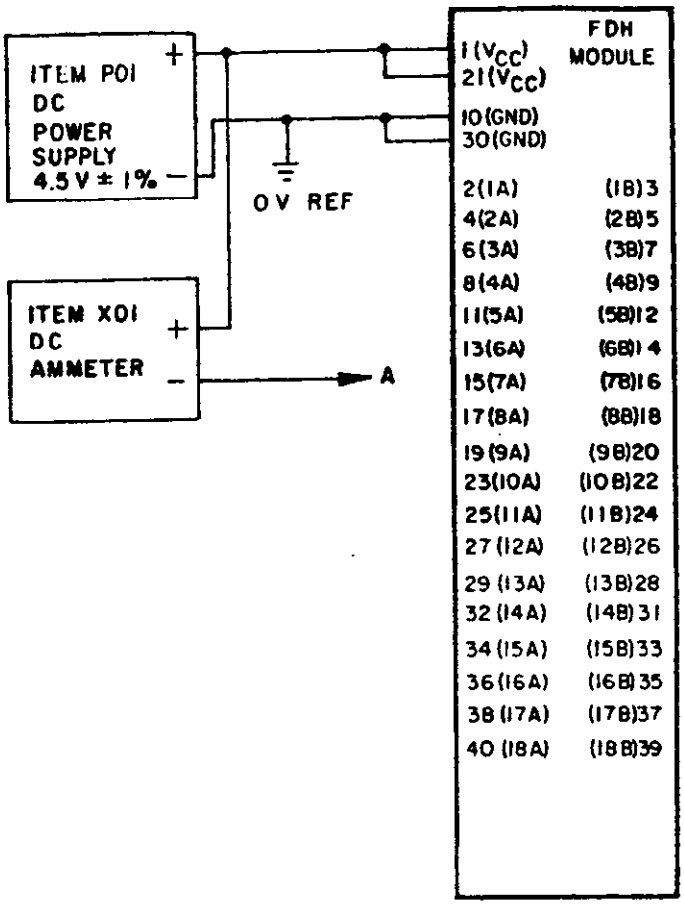
FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Static input current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
B1	Static input current	2	0	500	µA	6
B2		4				
B3		6				
B4		8				
B5		11				
B6		13				
B7		15				
B8		17				
B9		19				
B10		23				
B11		25				
B12		27				
B13		29				
B14		32				
B15		34				
B16		36				
B17		38				
B18		40				

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

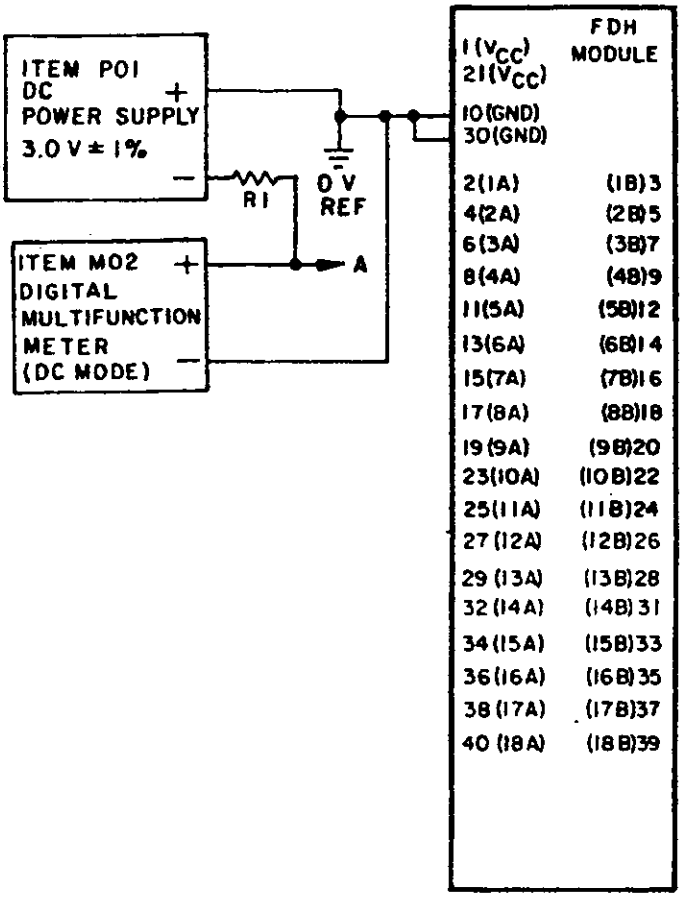
FIGURE 6. Static input current test circuit for table IV.

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TABLE V. Input clamp diode voltage test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C1	Input clamp diode voltage	2	-0.4	-1.5	V	7
C2		4				
C3		6				
C4		8				
C5		11				
C6		13				
C7		15				
C8		17				
C9		19				
C10		23				
C11		25				
C12		27				
C13		29				
C14		32				
C15		34				
C16		36				
C17		38				
C18		40				

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NOTE: R1 = 200Ω ±1%, 1/4 W.

FIGURE 7. Input clamp diode voltage test circuit for table V.

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TABLE VI. Output voltage (low-level and high-level) test.

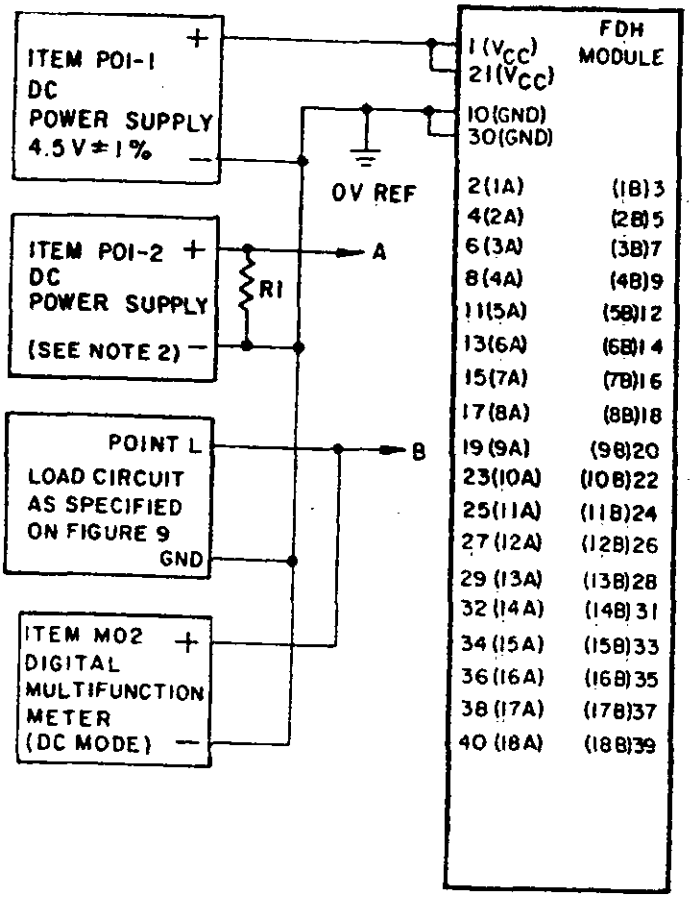
Test condition	Test	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
	D1	Output voltage, low level	2	3	0	450	mV	8
	D2		4	5				
	D3		6	7				
	D4		8	9				
	D5		11	12				
	D6		13	14				
	D7		15	16				
	D8		17	18				
	D9		19	20				
	D10		23	22				
	D11		25	24				
	D12		27	26				
	D13		29	28				
	D14		32	31				
	D15		34	33				
	D16		36	35				
	D17		38	37				
	D18		40	39				

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TABLE VI. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	Output voltage, high level	2	3	2.4	4.5	V	8
E2		4	5				
E3		6	7				
E4		8	9				
E5		11	12				
E6		13	14				
E7		15	16				
E8		17	18				
E9		19	20				
E10		23	22				
E11		25	24				
F12		27	26				
E13		29	28				
E14		32	31				
E15		34	33				
E16		36	35				
E17		38	37				
E18		40	39				

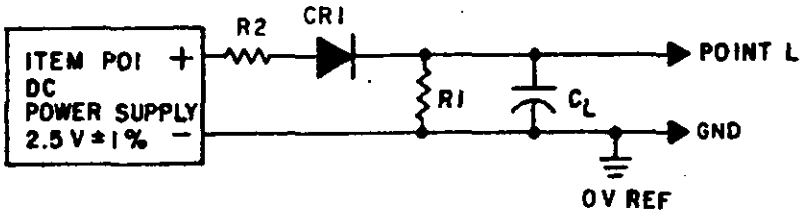
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NOTES:
1. R1 = 49.9Ω ±1%, 1 W.
2. Set item PO1-2 voltage as follows:
   - 2.0 V ±1% for test conditions D1-D18.
   - 0.8 V ±1% for test conditions F1-F18.

FIGURE 9. Output voltage (low-level and high-level) [illegible] for [illegible] VI.

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NOTES:
1. CR1 = 1N4150 or equivalent.
2. C_L = 130 ±5 pF (including probe and parasitic capacitance).
3. R1 = 4.02 kΩ ±1%, 1/8 W.
4. R2 = Select value for current flow of 22 mA ±1% out of point "L" with "L" to ground.
5. Capacitor C_L is not required for dc electrical conditions D or E.
6. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV peak.

FIGURE 9. FOH logic load circuit.

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TABLE VII. Input current (low-level and high-level) and output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	Input current, low level	Open	2	Open	1.0	2.2	mA	10
F2			4					
F3			6					
F4			8					
F5			11					
F6			13					
F7			15					
F8			17					
F9			19					
F10			23					
F11			25					
F12			27					
F13			29					
F14			32					
F15			34					
F16			36					
F17			38					
F18			40					

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

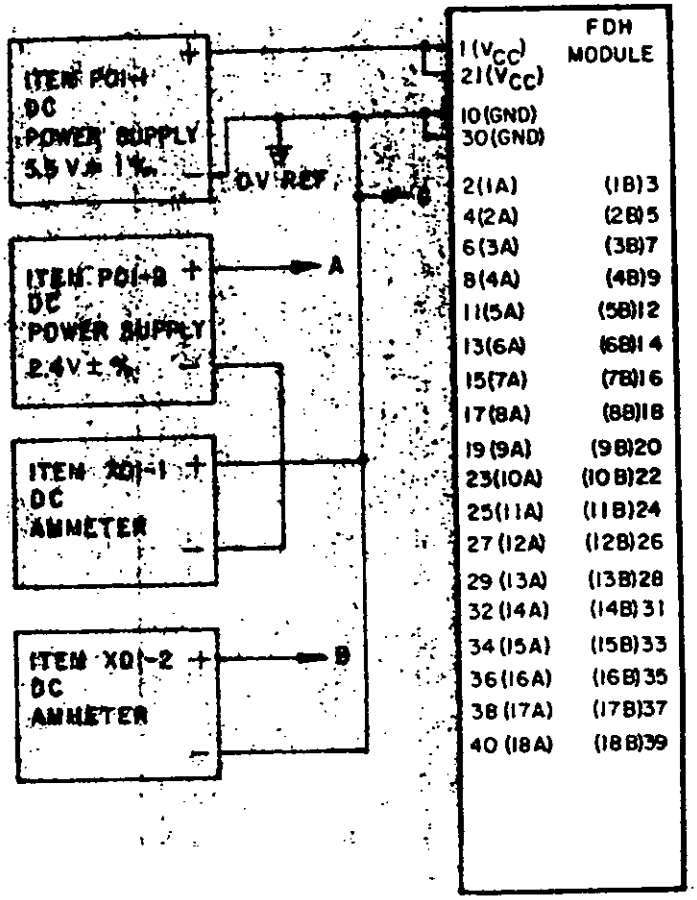
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	Input current, high level	2	Open	Open	0	60	µA	10
G2		4						
G3		6						
G4		8						
G5		11						
G6		13						
G7		15						
G8		17						
G9		19						
G10		23						
G11		25						
G12		27						
G13		29						
G14		32						
G15		34						
G16		36						
G17		38						
G18		40						

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	Output short circuit current	Open	3	2	40	100	mA	10
H2			5	4				
H3			7	6				
H4			9	8				
H5			12	11				
H6			14	13				
H7			16	15				
H8			18	17				
H9			20	19				
H10			22	23				
H11			24	25				
H12			26	27				
H13			28	29				
H14			31	32				
H15			33	34				
H16			35	36				
H17			37	38				
H18			39	40				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
2. With point A disconnected, item X01-1 should indicate zero current on the 1 µA range.
3. When performing output short circuit current test, only one output shall be shorted at a time, and the measurement taken within 5 seconds.

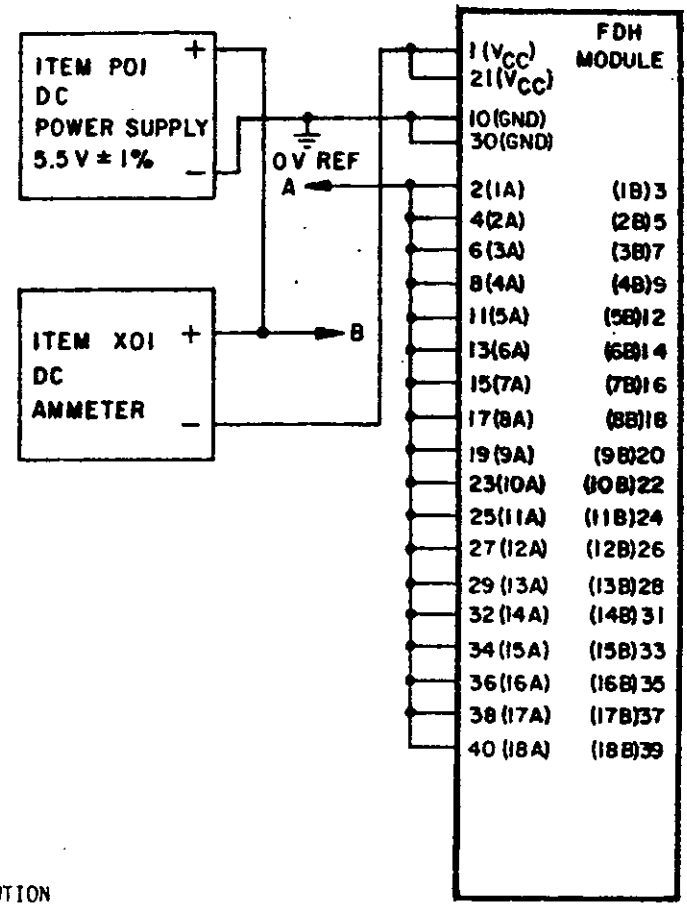
FIGURE 10. Input current (low-level and high-level) and output short circuit current test circuit for table VII.

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TABLE VIII. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
J1	Power supply current, low level	B	25	180	mA	11
J2	Power supply current, high level	Pin 10	10.0	75.6	mA	11

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 11. Power supply current (low-level and high-level) test circuit for table VIII.

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TABLE IX. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	1B Transition time (t_{THL})	2	3	3	20	ns	12,13
K2	2B	4	5				
K3	3B	6	7				
K4	4B	8	9				
K5	5B	11	12				
K6	6B	13	14				
K7	7B	15	16				
K8	8B	17	18				
K9	9B	19	20				
K10	10B	23	22				
K11	11B	25	24				
K12	12B	27	26				
K13	13B	29	28				
K14	14B	32	31				
K15	15B	34	33				
K16	16B	36	35				
K17	17B	38	37				
K18	18B	40	39				

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TABLE IX. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
1	1B Transition time (t_{TLH})	2	3	3	20	ns	12,13
2	2B	4	5				
3	3B	6	7				
4	4B	8	9				
5	5B	11	12				
6	6B	13	14				
7	7B	15	16				
8	8B	17	18				
9	9B	19	20				
10	10B	23	22				
11	11B	25	24				
12	12B	27	26				
13	13B	29	28				
14	14B	32	31				
15	15B	34	33				
16	16B	36	35				
17	17B	38	37				
18	18B	40	39				

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TABLE X. Propagation delay time (t_{PLH} and t_{PHL}) test.

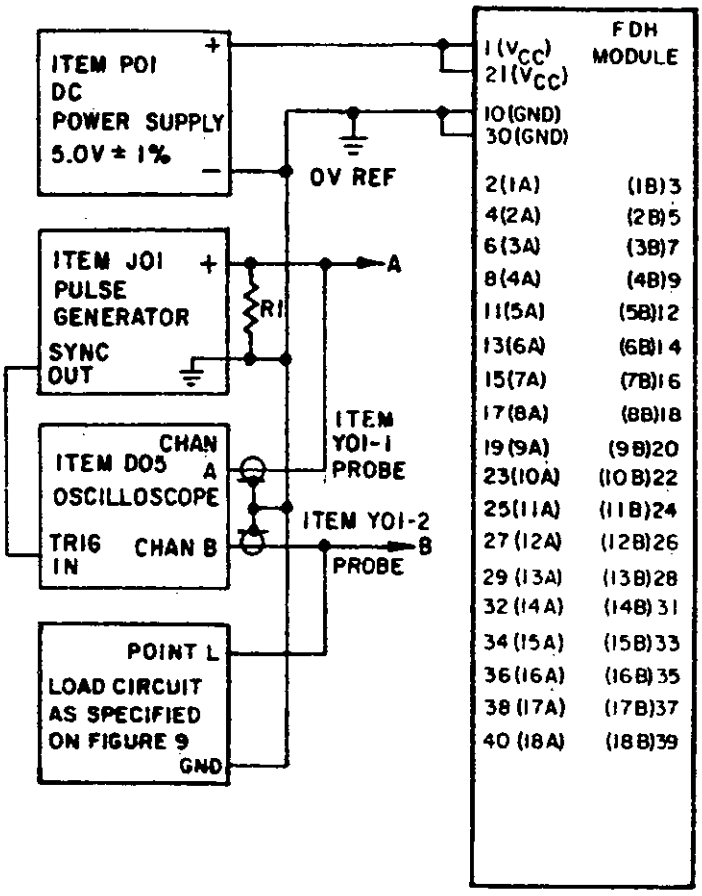
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	1B Propagation delay time (t_{PHL})	2	3	3	20	ns	12,13
M2	2B	4	5				
M3	3B	6	7				
M4	4B	8	9				
M5	5B	11	12				
M6	6B	13	14				
M7	7B	15	16				
M8	8B	17	18				
M9	9B	19	20				
M10	10B	23	22				
M11	11B	25	24				
M12	12B	27	26				
M13	13B	29	28				
M14	14B	32	31				
M15	15B	34	33				
M16	16B	36	35				
M17	17B	38	37				
M18	18B	40	39				

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TABLE X. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	1B Propagation delay time (t_{PHL})	2	3	3	20	ns	12,13
N2	2B	4	5				
N3	3B	6	7				
N4	4B	8	9				
N5	5B	11	12				
N6	6B	13	14				
N7	7B	15	16				
N8	8B	17	18				
N9	9B	19	20				
N10	10B	23	22				
N11	11B	25	24				
N12	12B	27	26				
N13	13B	29	28				
N14	14B	32	31				
N15	15B	34	33				
N16	16B	36	35				
N17	17B	38	37				
N18	18B	40	39				

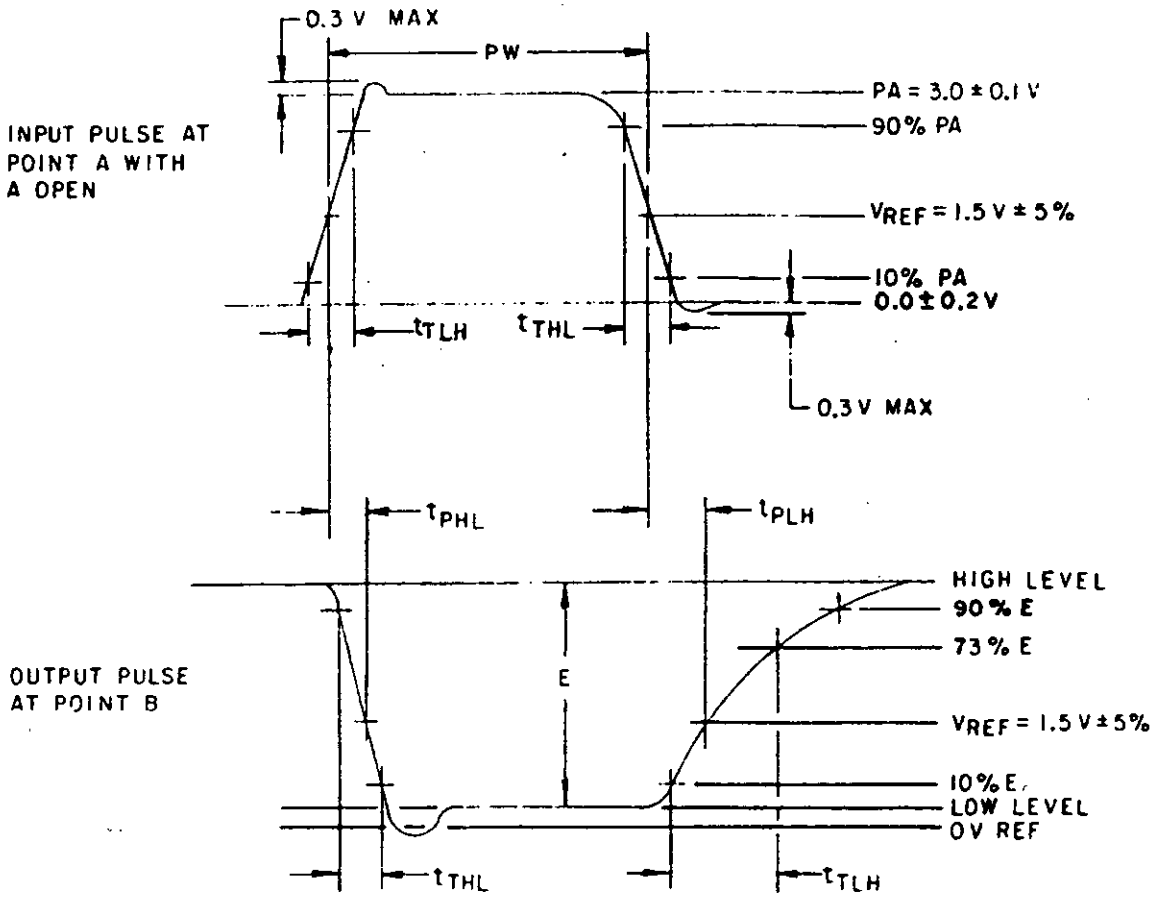
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NOTE: R1 = 49.9 Ω ±1%, 1 W.

FIGURE 12. Transition (t_{TLH} and t_{THL}) and propagation delay (t_{PLH} and t_{PHL}) time test circuit for tables IV and V.

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NOTES:
1. Input pulse parameters:
   PW = 500 ns ±10%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PRR = 1.0 MHz ±10%.
2. Before making the measurement, verify that the output pulse high level is ≥ 2.4 V and the low level is ≤ 0.45 V.

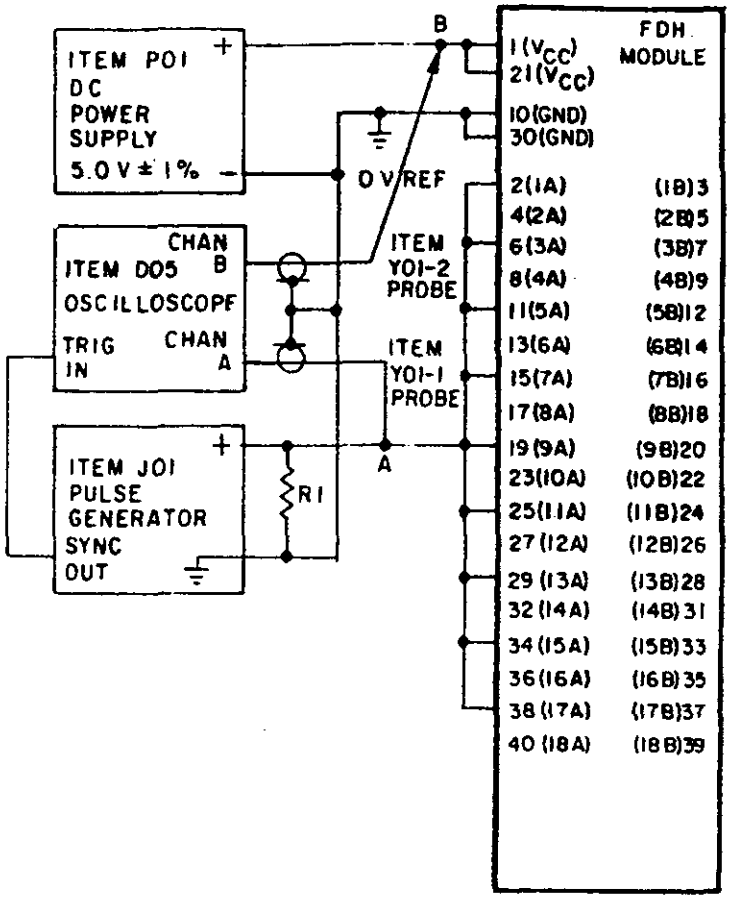
FIGURE 13. Transition (t_{TLH} and t_{THL}) and propagation delay (t_{PLH} and t_{PHL}) time test circuit waveforms for tables IX and X.

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TABLE XI. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
P	Power supply transient amplitude	0	400	mV	14,15

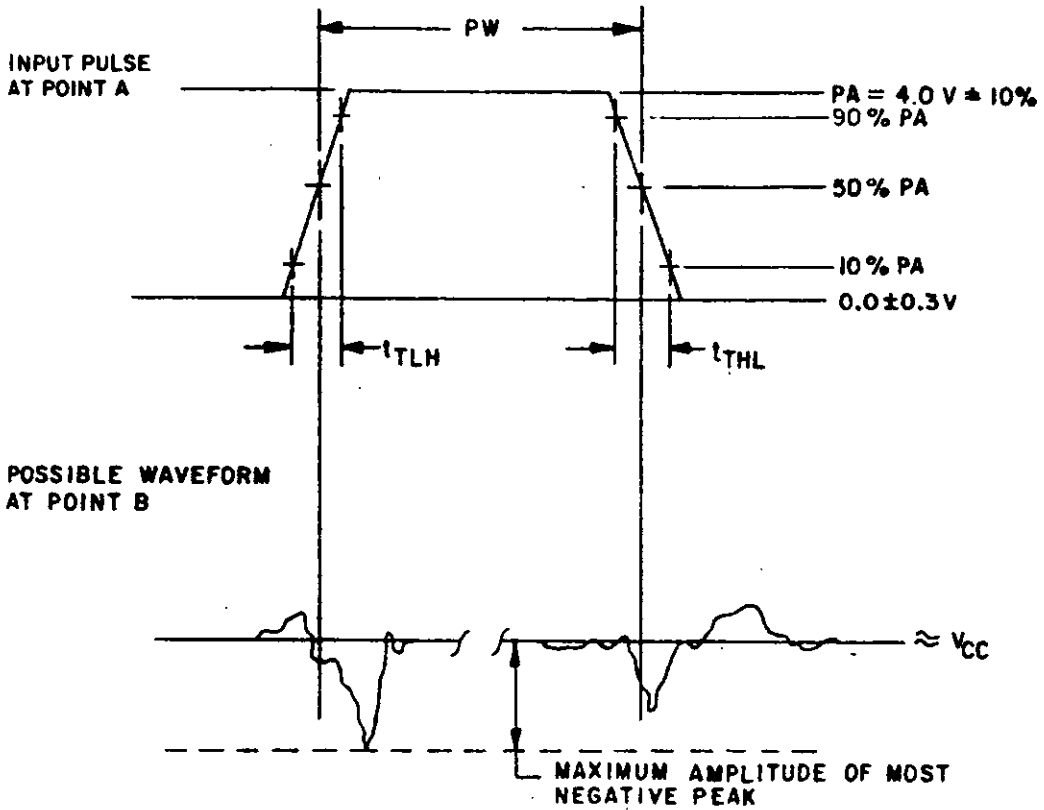
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NOTE: R1 = 49.9 Ω ±1%, 1 W.

FIGURE 14. Power supply transient amplitude test circuit for table XI.

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NOTE: Input pulse parameters:
PW = 3 µs ±10%.
$t_{TLH} = t_{THL}$ = 10 ns ±10% (verified at each input pin).
PRR = 100 kHz ±10 .

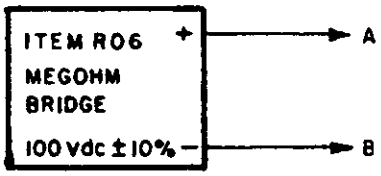
FIGURE 15. Power supply transient amplitude test circuit waveforms for table XI.

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TABLE XII. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q	Isolation	All pins	Frame	10	∞	MΩ	16

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	FDH MODULE
1 (V_{CC})	
21 (V_{CC})	
10 (GND)	
30 (GND)	
2 (1A)	(1B) 3
4 (2A)	(2B) 5
6 (3A)	(3B) 7
8 (4A)	(4B) 9
11 (5A)	(5B) 12
13 (6A)	(6B) 14
15 (7A)	(7B) 16
17 (8A)	(8B) 18
19 (9A)	(9B) 20
23 (10A)	(10B) 22
25 (11A)	(11B) 24
27 (12A)	(12B) 26
29 (13A)	(13B) 28
32 (14A)	(14B) 31
34 (15A)	(15B) 33
36 (16A)	(16B) 35
38 (17A)	(17B) 37
40 (18A)	(18B) 39

WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 16. Isolation test circuit for table XII.

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TABLE XIII. Output crosstalk (low-level and high- level) test.

Test condition		Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R1	1B	Output crosstalk, low level	2	4,6,8,11,13,15, 17,19,23,25,27, 29,32,34,36, 38,40	3	0	400	mV	17,18
R2	2B		4	2,6,8,11,13,15, 17,19,23,25,27, 29,32,34,36, 38,40	5				
R3	3B		6	2,4,8,11,13,15, 17,19,23,25,27, 29,32,34,36, 38,40	7				
R4	4B		8	2,4,6,11,13,15, 17,19,23,25,27, 29,32,34,36, 38,40	9				
R5	5B		11	2,4,6,8,13,15, 17,19,23,25,27, 29,32,34,36, 38,40	12				
R6	6B		13	2,4,6,8,11,15, 17,19,23,25,27, 29,32,34,36,38, 40	14				
R7	7B		15	2,4,6,8,11,17, 19,23,25,27,29, 32,34,36,38,40	16				
R8	8B		17	2,4,6,8,11,15, 19,23,25,27,29, 32,34,36,38,40	18				
R9	9B		19	2,4,6,8,11,15, 17,23,25,27,29, 32,34,36,38,40	20				

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Check the source to verify that this is the current version before use.

TABLE XIII. Output crosstalk (low-level and high-level) test - Continued.

Test condition		Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R	10	10B Output crosstalk, low level	23	2,4,6,8,11,13, 15,17,19,25,27, 29,32,34,36, 38,40	22	0	400	mV	17,18
R	11	11B	25	2,4,6,8,11,13, 15,17,19,23,27, 29,32,34,36, 38,40	24				
R	12	12B	27	2,4,6,8,11,13, 15,17,19,23,25, 29,32,34,36, 38,40	26				
R	13	13B	29	2,4,6,8,11,13, 15,17,19,23,25, 27,32,34,36, 38,40	28				
R	14	14B	32	2,4,6,8,11,13, 15,17,19,23,25, 27,29,34,36, 38,40	31				
R	15	15B	34	2,4,6,8,11,13, 15,17,19,23,25, 27,29,32,36, 38,40	33				
R	16	16B	36	2,4,6,8,11,13, 15,17,19,23,25, 27,29,32,34, 38,40	35				
R	17	17B	38	2,4,6,8,11,13, 15,17,19,23,25, 27,29,32,34,36, 40	37				
R	18	18B	40	2,4,6,8,11,13, 15,17,19,23,25, 27,29,32,34, 36,38	39				

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Check the source to verify that this is the current version before use.

TABLE XIII. Output crosstalk (low-level and high-level) test - Continued.

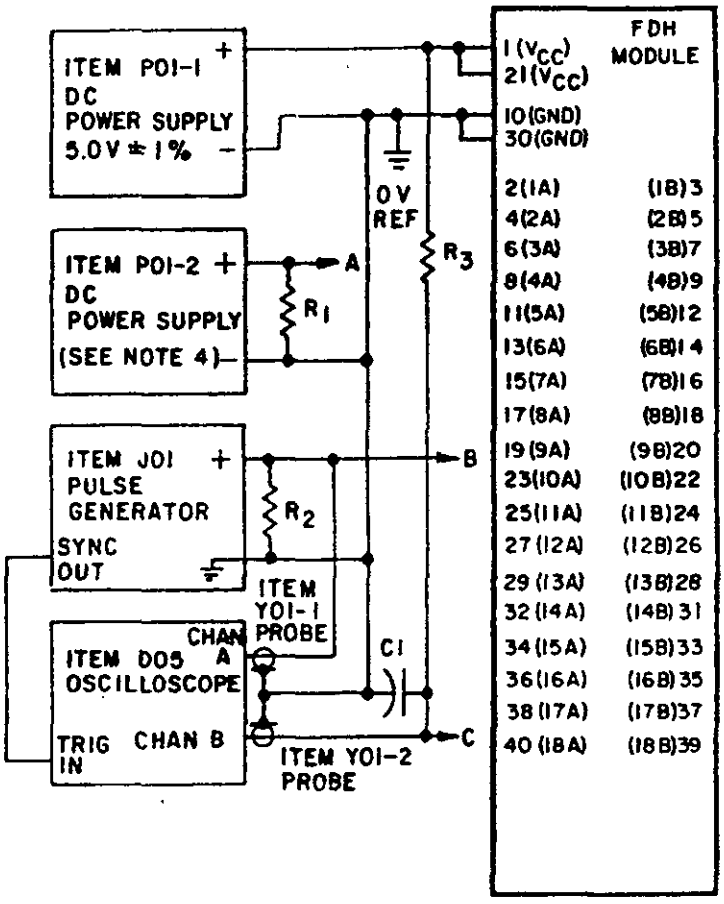
Test condition		Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
S1	1B	Output crosstalk, high level	2	4,6,8,11,13,15, 17,19,23,25,27, 29,32,34,36, 38,40	3	0	400	mV	17,18
S2	2B		4	2,6,8,11,13,15, 17,19,23,25,27, 29,32,34,36, 38,40	5				
S3	3B		6	2,4,8,11,13,15, 17,19,23,25,27, 29,32,34,36, 38,40	7				
S4	4B		8	2,4,6,11,13,15, 17,19,23,25,27, 29,32,34,36, 38,40	9				
S5	5B		11	2,4,6,8,13,15, 17,19,23,25,27, 29,32,34,36, 38,40	12				
S6	6B		13	2,4,6,8,11,15, 17,19,23,25,27, 29,32,34,36, 38,40	14				
S7	7B		15	2,4,6,8,11,13, 17,19,23,25,27, 29,32,34,36, 38,40	16				
S8	8B		17	2,4,6,8,11,13, 15,19,23,25,27, 29,32,34,36, 38,40	18				
S9	9B		19	2,4,6,8,11,13, 15,17,23,25,27, 29,32,34,36, 38,40	20				

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TABLE XIII. Output crosstalk (low-level and high-level) test - Continued.

Test condition		Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
S10	10B	Output crosstalk, high level	23	2,4,6,8,11,13,15,17,19,25,27,29,32,34,36,38,40	22	0	400	mV	17,18
S11	11B		25	2,4,6,8,11,13,15,17,19,23,27,29,32,34,36,38,40	24				
S12	12B		27	2,4,6,8,11,13,15,17,19,23,25,29,32,34,36,38,40	26				
S13	13B		29	2,4,6,8,11,13,15,17,19,23,25,27,32,34,36,38,40	28				
S14	14B		32	2,4,6,8,11,13,15,17,19,23,25,27,29,34,36,38,40	31				
S15	15B		34	2,4,6,8,11,13,15,17,19,23,25,27,29,32,36,38,40	33				
S16	16B		36	2,4,6,8,11,13,15,17,19,23,25,27,29,32,34,38,40	35				
S17	17B		38	2,4,6,8,11,13,15,17,19,23,25,27,29,32,34,36,40	37				
S18	18B		40	2,4,6,8,11,13,15,17,19,23,25,27,29,32,34,36,38	39				

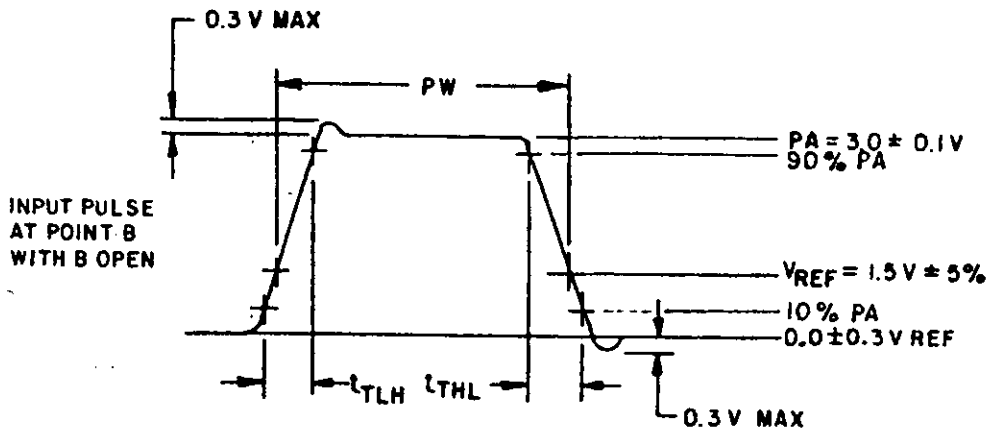
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Check the source to verify that this is the current version before use.

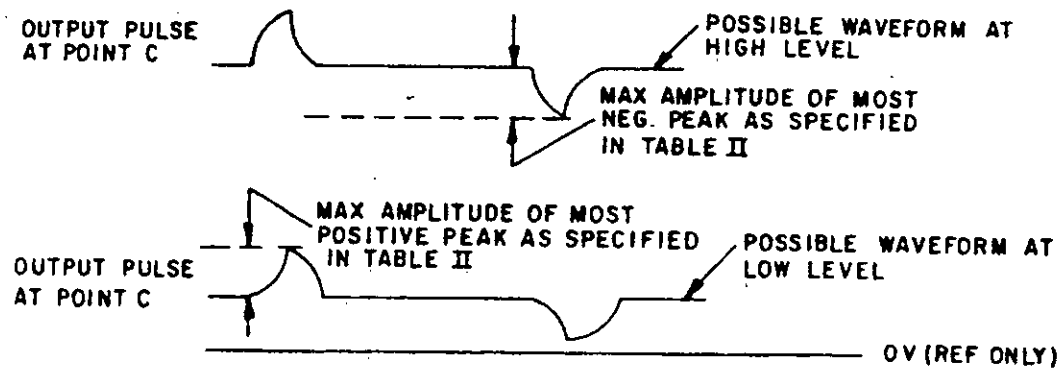


NOTES:
1. R1 - R2 = 49.9Ω ±1%, 1 W.
2. R3 = 243Ω ±1%, 1/4 W.
3. C1 = 30 ±1 pF (includes probe capacitance).
4. Set item POI-2 voltage as follows:
   V_{IN} = 2.4 V ±1% for test conditions R1-R18.
   V_{IN} = 0.4 V ±1% for test conditions S1-S18.
5. Only the output under test shall be loaded. The lead length from pin 1 to each load resistor and output including the length of the resistor body shall not exceed 4.5 inches (114.30 mm).

FIGURE 12. Output rise (fall) time and propagation delay test circuit for table VIII.

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NOTE: Input pulse parameters:
PW = 200 ±20 ns.
$t_{TLH} = t_{THL}$ = 25 ±5 ns.
PRR = 1.0 MHz ±10%.

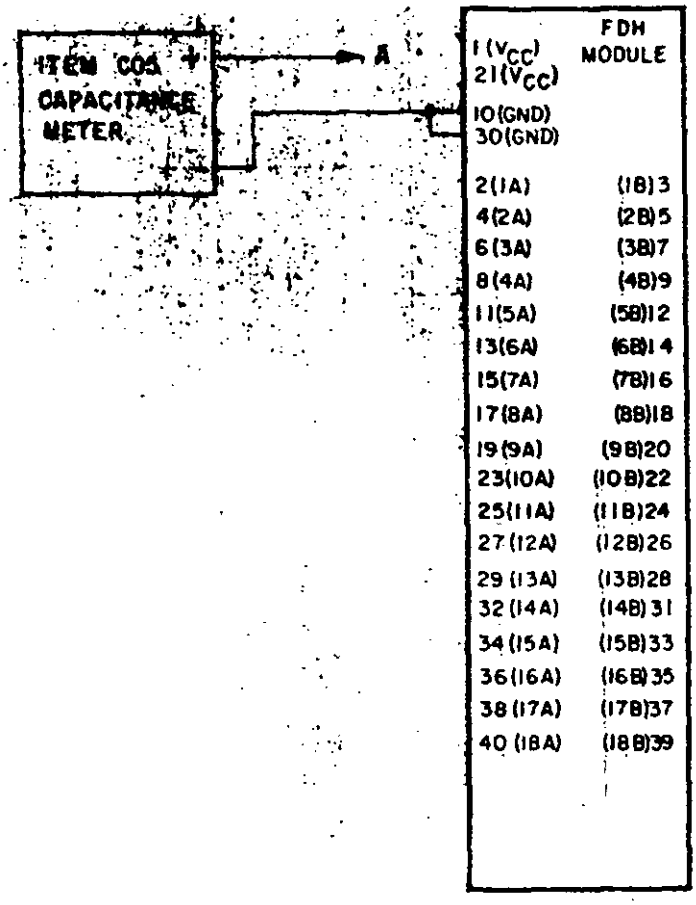
FIGURE 18. Output crosstalk (low-level and high-level) test circuit waveforms for table XIII.

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TABLE XIV. Input capacitance test.

Test condition		Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
T1	1A	Input capacitance	2	2	12	pF	19
T2	2A		4				
T3	3A		6				
T4	4A		8				
T5	5A		11				
T6	6A		13				
T7	7A		15				
T8	8A		17				
T9	9A		19				
T10	10A		23				
T11	11A		25				
T12	12A		27				
T13	13A		29				
T14	14A		32				
T15	15A		34				
T16	16A		36				
T17	17A		38				
T18	18A		40				

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pins 10 and 30 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Zero out the effects of the test leads prior to connecting to the module.

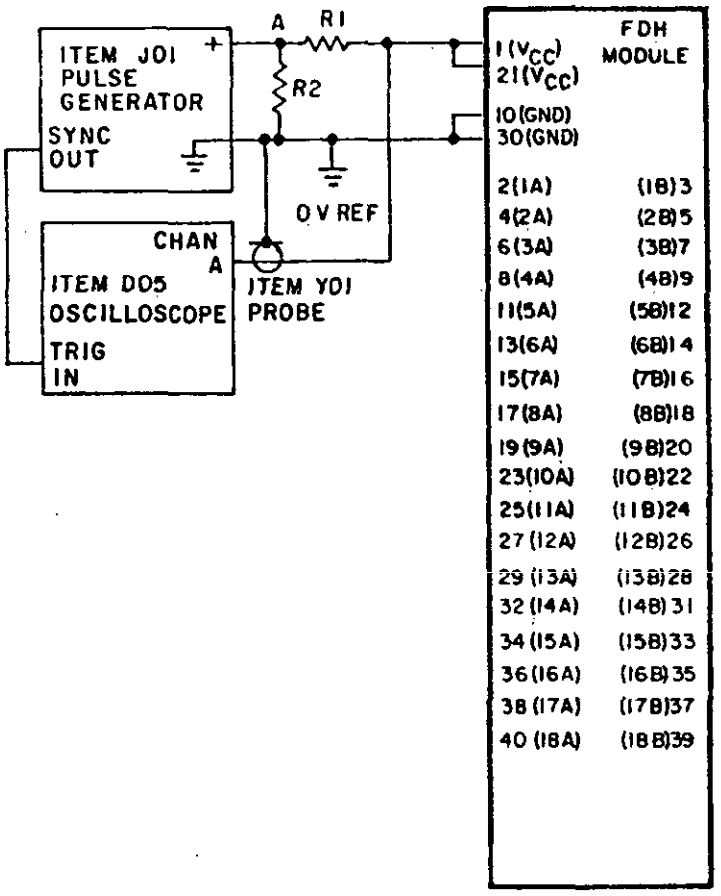
FIGURE 19. Input capacitance test circuit for table XIV.

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TABLE XV. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
3	Power supply transient rise time	2.5	7.5	µs	20,21

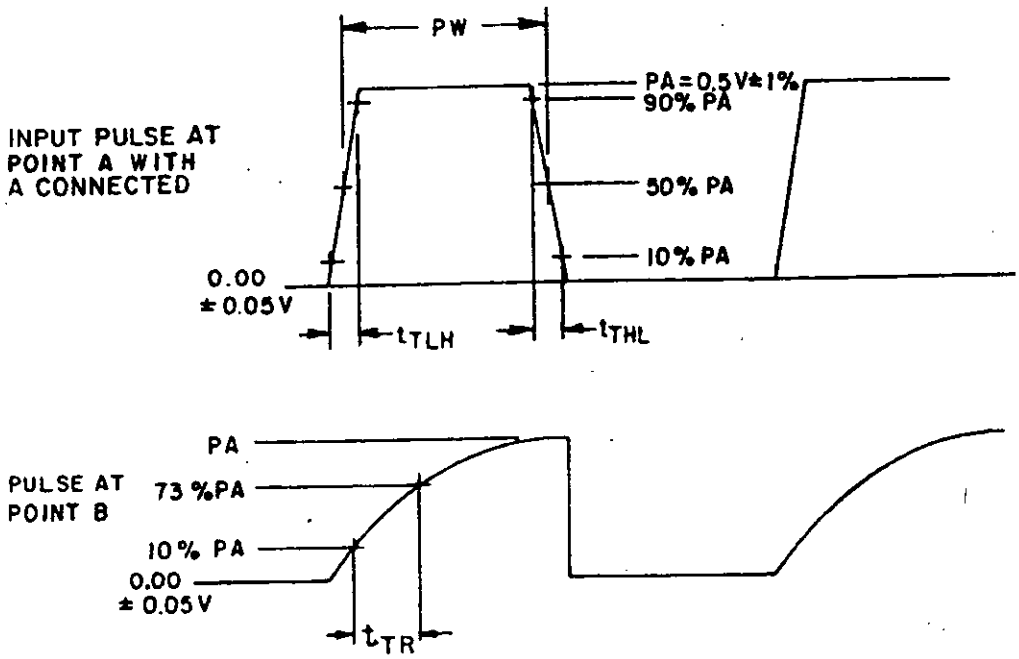
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NOTES:
1. R1 = 412Ω ±1%, 1/8 W.
2. R2 = 49.9Ω ±1%, 1 W.

FIGURE 20. Power supply transient rise time test circuit for table IV.

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NOTES:
1. Input pulse parameters:
   PW = 30 µs ±10%.
   $t_{TLH} = t_{THL}$ = 50 ns ±10%.
   PRP = 60 µs ±10%.
2. Measure transient rise time (t_{TR}) at point B.

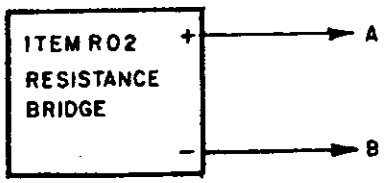
FIGURE 21. Power supply transient rise time test circuit waveforms for table XV.

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TABLE XVI. Continuity test.

Test condition	Parameter	Connect		Initial limit 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V1	V_{CC} pin continuity	1	21	0	1	Ω	22
V2	Ground pin continuity	30	10	0	1	Ω	22

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	FDH MODULE
1(V_{CC}) 21(V_{CC})	
10(GND) 30(GND)	
2(1A)	(1B)3
4(2A)	(2B)5
6(3A)	(3B)7
8(4A)	(4B)9
11(5A)	(5B)12
13(6A)	(6B)14
15(7A)	(7B)16
17(8A)	(8B)18
19(9A)	(9B)20
23(10A)	(10B)22
25(11A)	(11B)24
27(12A)	(12B)26
29(13A)	(13B)28
32(14A)	(14B)31
34(15A)	(15B)33
36(16A)	(16B)35
38(17A)	(17B)37
40(18A)	(18B)39

FIGURE 22. Continuity test circuit for table XVI.

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